

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – SHORT ANSWER TEST

Course Code:	CSA16	Course Title:	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL1, BL2)	Assessment:	Assessment Tool 1 – Short Answer Test
Weightage:	10% of CIA	Total Marks:	100 (5 Questions × 20 Marks)
CO1 Statement:	<i>Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2</i>		
Student Name:	M. Megharadh	Reg. No.:	192325026
Section / Batch:	AI & ML	Date:	15/7/26

Instructions to Students

- Answer all questions. Each question carries 20 Marks. Total: 100 Marks.
- Clearly show all requirements, process, operations.
- Use appropriate models, schema represented scenario wise
- Justify each step in different dimension specified and measures clearly.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Short Answer Test	Architecture of Data Warehouse, ETL, OLAP	Q1 – Q5	Q1 –20 Q2 –20 Q3 –20 Q4 –20 Q5 –20
GRAND TOTAL:			100 Marks

SECTION A – SHORT ANSWER TEST

1. Suppose that a data warehouse for City Hospital consists of the following four dimensions: patient, doctor, department, and time, and two measures visit_count and avg_charge. At the lowest conceptual level, avg_charge stores the actual consultation fee charged for a patient visit. At higher conceptual levels, it stores the average fee.

(a) Draw a snowflake schema diagram for the data warehouse.

(b) Starting with the base cuboid [patient, doctor, department, time], what specific OLAP operations should be performed to list the average consultation fee department-wise for each year?

(c) If each dimension has four levels (including all), how many cuboids will this cube contain (including base and apex cuboids)?

2. A smart city project is collecting data from multiple sources, including:

- Traffic sensors for vehicle counts and congestion levels.
- IoT devices for monitoring air quality, noise levels, and energy consumption.
- Public transportation systems for ridership data and punctuality.
- Citizen feedback from a mobile app.

Questions:

Fact and Dimension Design:

- Design a star schema to analyze traffic congestion patterns, energy consumption trends, and public transportation efficiency.
- What measures and dimensions would you include? Justify your design.

3. Suppose that a banking data warehouse contains two fact tables: Transaction_Fact and Loan_Fact, sharing dimensions customer, branch, time, and account_type.

- Transaction_Fact → transaction_count, amount
- Loan_Fact → loan_amount, interest

(a) Draw a galaxy schema diagram.

(b) Identify OLAP operations to obtain branch-wise average transaction amount and loan amount yearly.

(c) If each dimension has four levels, calculate total cuboids.

4. A manufacturing company monitors product quality and production efficiency. The data warehouse contains:

- Dimensions:
 - Product: Category, type, and batch.
 - Factory: Location, department, and line number.
 - Time: Year, quarter, month, and week.
 - Supplier: Region, rating, and contract type.
- Measures:
 - Defect rate.
 - Production units.
 - Supplier reliability percentage.

Questions:

- Roll-up: Summarize defect rates by product category for the current year across all factories.
- Drill-down: Focus on a specific factory to analyze defect rates by department and product type.
- Slice: Isolate production unit data for suppliers with a rating of 4+ across all regions.
- Dice: Analyze defect rates and supplier reliability for selected product categories and factory locations.

5. A retail chain manages inventory and sales data across multiple stores. The data warehouse contains:

- Dimensions:
 - Product: Category, subcategory, and SKU.
 - Store: Region, city, and store ID.
 - Time: Year, quarter, month, and day.
 - Customer: Membership tier, age group, and gender.
- Measures:
 - Inventory turnover.
 - Total sales revenue.
 - Units sold.

Questions:

1. Roll-up: Summarize inventory turnover by product category across all stores for the last quarter.
2. Drill-down: Focus on sales data for electronics in a specific region, analysing by subcategory and SKU.
3. Slice: Isolate sales revenue data for male customers in the Bronze membership tier across all stores.
4. Dice: Analyze sales revenue and units sold for selected cities and product categories during specific time periods.

Code of Conduct

I, M. Megha (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M. Megha

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Marks Evaluation:

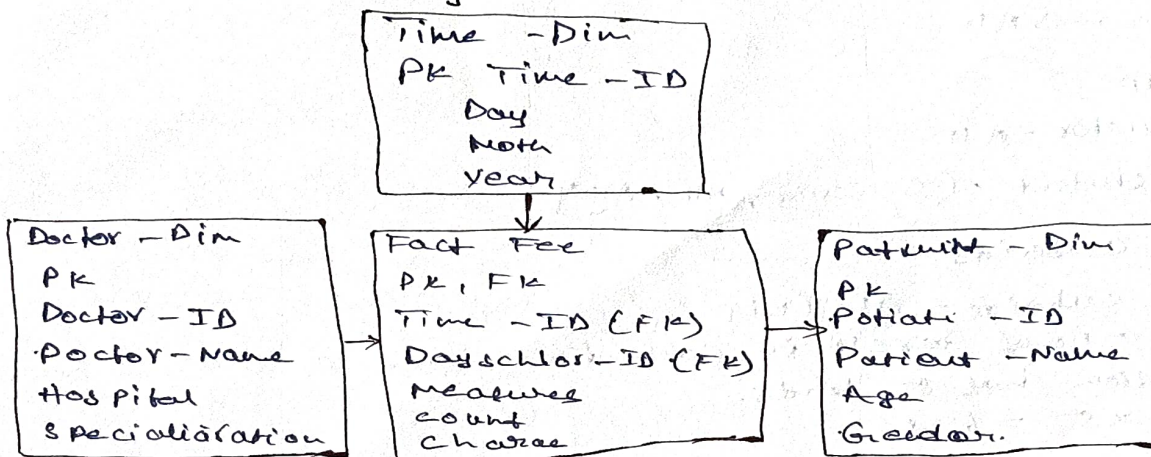
Question No.	Understanding of Concepts (25%)	Explanation (25%)	Application (20%)	Organization (15%)	Accuracy (10%)	Clarity (5%)	Total Marks
Q1 (20 Marks)							
Q2 (20 Marks)							
Q3 (20 Marks)							
Q4 (20 Marks)							
Q5 (20 Marks)							
Overall Total (100)							

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Data warehouse Doctor - Patient visits

a) Three popular schemas used in data warehouse are

- 1) Star schema
 - 2) snow flake schema
 - 3) Fact constellation schema
- b) Star schema diagram.



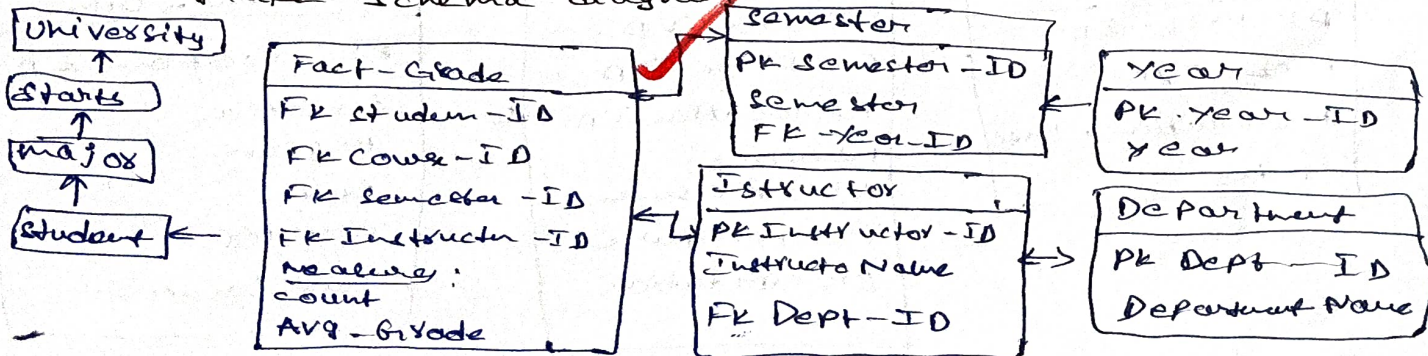
c) SQL query

```

SELECT
  doctor
  Patient
  sum (count) As Total - visits
  sum (charges) As Total - charges
FROM Fee
GROUP BY doctor, Patient
  
```

2) Big university data warehouse

a) snow flake schema diagram



5) OLAP operation

Starting Cuboid: [Student, course, semester, Instructor]

operations:

1. Slice.
* select only CS course
course = 'CS'
2. Roll-up
* semester → All
3. Roll-up
* Instructor → All

Result: [Student, CS, course, All, All]

c) No. of cuboids

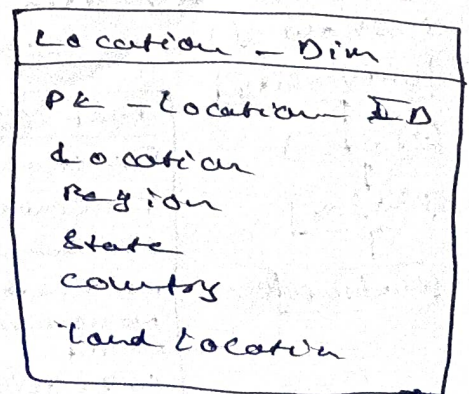
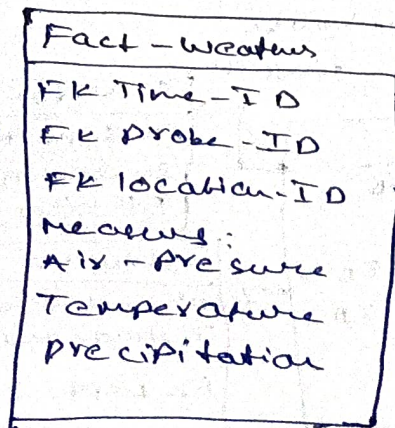
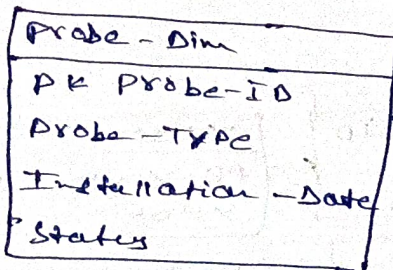
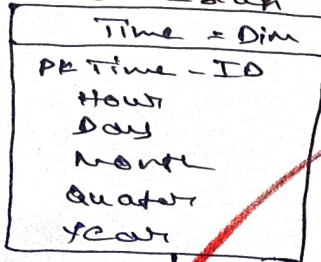
Formula = cuboids = $\prod (L_i)$

where L_i = no. of levels in each dimension.
Each dimension has 5 levels.

$$5 \times 5 \times 5 \times 5 = 625$$

3) Regional Weather Bureau Data Warehouse.

Star-schema design



Advantage:-

- * Analyze temperature by region / year
- * Compare land vs ocean weather
- * Perform OLAP Operation efficiently.

4) Data cube problem

Given n dimension.

Each dimension has p distinct value.
No hierarchies.

- Max. Cells in Base cuboid
Every combination exists $[p^n]$.
- Minimum cells in Base cuboid
only one non-empty cell $[1]$.
- Max cells in entire cube
each dimension contributes $(p+1)$
 $\therefore (p+1)^n$.
- Min. cells in entire cube

If only one base cell exists.

Every cuboid contributes one aggregate cell

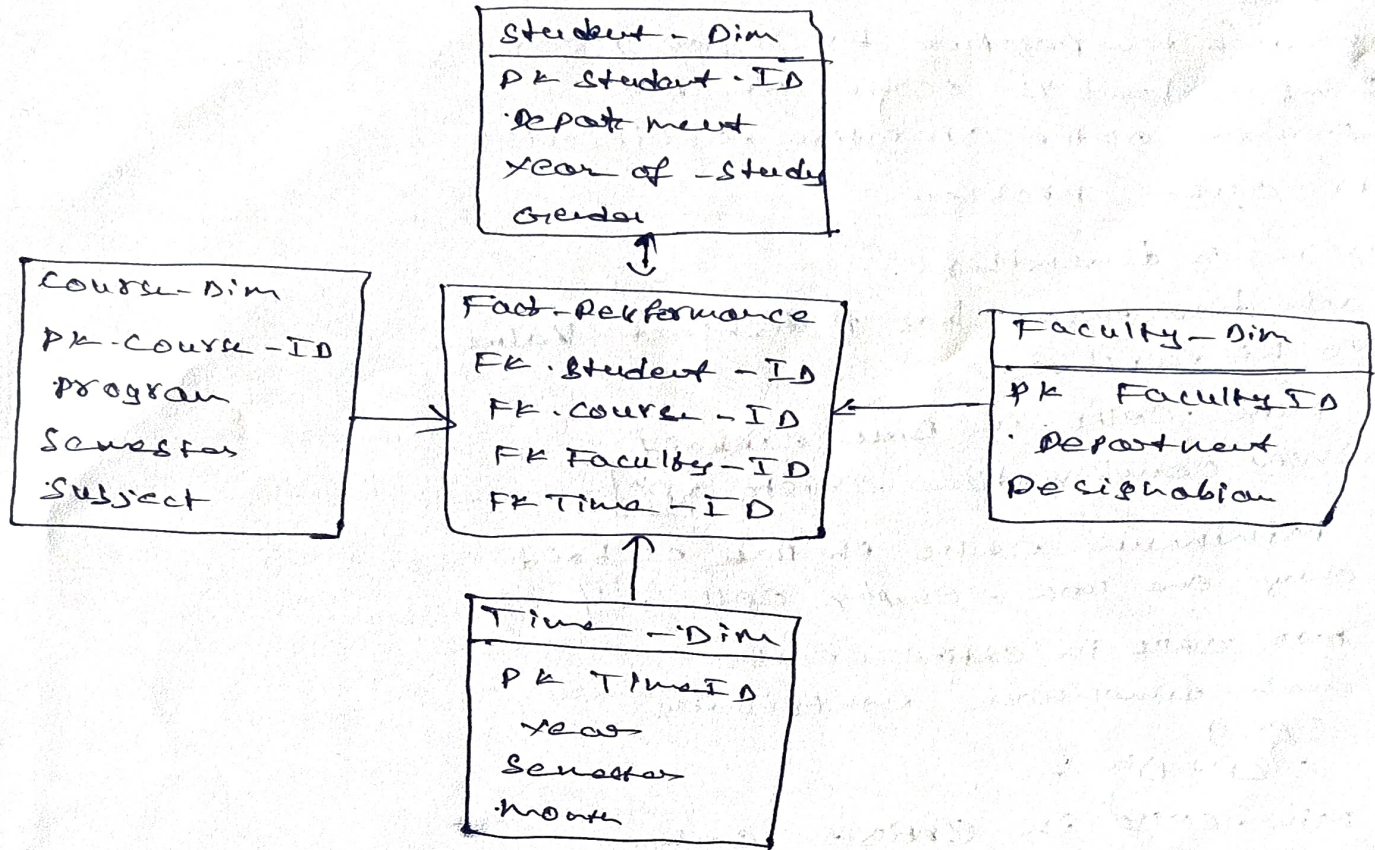
No. of cuboids $[2^n]$

5) University Academic Performance Data were loaded.

OLAP Operation

Step	Operation	Description	Result
1.	Roll-up	Student $\rightarrow$ department	[Department, Course, Faculty,]
2.	Roll-up	Time: Month $\rightarrow$ Semester $\rightarrow$ year	[Department, Course, Faculty, year]
3.	Roll-up	Course $\rightarrow$ All	[Department, All, Faculty,]
4.	Roll-up	Faculty $\rightarrow$ All	[Department, All, All, year]
<u>Result:</u> [Department, All, All, year]			

a) Star-Schema design



c) Total no. of cuboids

Formula : $\text{Cuboids} = L_1 \times L_2 \times L_3 \times L_4$

Given, Student = 4 levels

Course = 3 levels

Faculty = 3 levels

Time = 5 Levels

$$4 \times 3 \times 3 \times 5 = 180$$

$\therefore$ 180 Cuboids